

# BIOLOGY 30

## Unit 1 — Nervous and Endocrine Systems Study Guide

### 1 Unit / Module Objectives

| Topic Area             | Diploma-Level Objective                                                                        |
|------------------------|------------------------------------------------------------------------------------------------|
| Neuron Structure       | Analyze neuron structure and explain how electrochemical gradients generate action potentials. |
| Action Potentials      | Explain depolarization, repolarization, and refractory periods in nerve impulse transmission.  |
| Synaptic Transmission  | Explain neurotransmitter release and receptor activation across synapses.                      |
| Central Nervous System | Interpret functions of cerebrum, cerebellum, hypothalamus, and medulla.                        |
| Reflex Arcs            | Trace reflex arc pathways including sensory neurons, interneurons, and motor neurons.          |
| Vision                 | Explain how rods and cones convert light stimuli into neural signals.                          |
| Hearing and Balance    | Explain how the cochlea and semicircular canals detect sound and equilibrium.                  |
| Endocrine Glands       | Identify endocrine glands and analyze hormone functions.                                       |
| Hormonal Feedback      | Interpret negative feedback mechanisms regulating hormone secretion.                           |
| Homeostasis            | Evaluate how nervous and endocrine systems coordinate physiological responses.                 |

### 2 Student Self-Verification Checklist

| Topic             | Self-Verification Statement                                                                      |
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| Neuron physiology | <input type="checkbox"/> I can explain sodium and potassium movement during an action potential. |

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| <b>Synapses</b>           | <input type="checkbox"/> I can describe how neurotransmitters trigger a postsynaptic response.          |
| <b>Brain anatomy</b>      | <input type="checkbox"/> I can identify functions of major brain structures.                            |
| <b>Reflex arcs</b>        | <input type="checkbox"/> I can trace receptor → sensory neuron → interneuron → motor neuron → effector. |
| <b>Vision</b>             | <input type="checkbox"/> I can explain how rods and cones respond to different light intensities.       |
| <b>Hearing</b>            | <input type="checkbox"/> I can explain how hair cells convert vibrations into nerve impulses.           |
| <b>Balance</b>            | <input type="checkbox"/> I can describe how semicircular canals detect rotational movement.             |
| <b>Hormones</b>           | <input type="checkbox"/> I can identify endocrine glands and their major hormones.                      |
| <b>Glucose regulation</b> | <input type="checkbox"/> I can explain insulin and glucagon regulation of blood glucose.                |
| <b>Homeostasis</b>        | <input type="checkbox"/> I can interpret negative feedback loops controlling hormones.                  |

### **3 Diploma-Level Practice Questions with Fully Worked Solutions**

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| <b>Question</b>                    | A neuron changes from $-70$ mV to $+30$ mV. Explain the ionic mechanism responsible.                                                                                                                                 |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                                                                                             |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                                                                                      |
| <b>Step 3 — Explicit Reasoning</b> | Sodium channels open allowing $\text{Na}^+$ ions to diffuse into the neuron down their electrochemical gradient. This influx of positive charge depolarizes the membrane until the potential reaches about $+30$ mV. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                                                                                            |
| <b>Step 5 — Final Answer</b>       | Sodium channels open allowing $\text{Na}^+$ ions to diffuse into the neuron down their electrochemical gradient. This influx of positive charge depolarizes the membrane until the potential reaches about $+30$ mV. |

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| <b>Common Error</b>  | Students often confuse neural impulses with hormonal responses.  |
| <b>Exam Strategy</b> | Identify stimulus → pathway → effector → physiological response. |

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| <b>Question</b>                    | Explain why myelinated neurons conduct impulses faster.                                                                                                |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                               |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                        |
| <b>Step 3 — Explicit Reasoning</b> | Myelin insulates the axon and forces the action potential to jump between nodes of Ranvier. This saltatory conduction greatly increases impulse speed. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                              |
| <b>Step 5 — Final Answer</b>       | Myelin insulates the axon and forces the action potential to jump between nodes of Ranvier. This saltatory conduction greatly increases impulse speed. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                                                        |
| <b>Exam Strategy</b>               | Identify stimulus → pathway → effector → physiological response.                                                                                       |

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| <b>Question</b>                    | Trace the pathway of the withdrawal reflex after touching a hot object.                                                                                                                          |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                                                                         |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                                                                  |
| <b>Step 3 — Explicit Reasoning</b> | Heat receptors activate a sensory neuron which sends an impulse to the spinal cord. An interneuron relays the signal to a motor neuron which stimulates muscle contraction withdrawing the limb. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                                                                        |
| <b>Step 5 — Final Answer</b>       | Heat receptors activate a sensory neuron which sends an impulse to the spinal cord. An interneuron relays the signal to a motor neuron which stimulates muscle contraction withdrawing the limb. |

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| <b>Common Error</b>  | Students often confuse neural impulses with hormonal responses.  |
| <b>Exam Strategy</b> | Identify stimulus → pathway → effector → physiological response. |

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| <b>Question</b>                    | Explain the functional difference between rods and cones.                                                                                              |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                               |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                        |
| <b>Step 3 — Explicit Reasoning</b> | Rods detect low light and allow night vision but do not detect colour. Cones detect colour and function best in bright light, especially in the fovea. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                              |
| <b>Step 5 — Final Answer</b>       | Rods detect low light and allow night vision but do not detect colour. Cones detect colour and function best in bright light, especially in the fovea. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                                                        |
| <b>Exam Strategy</b>               | Identify stimulus → pathway → effector → physiological response.                                                                                       |

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| <b>Question</b>                    | Explain how the cochlea converts sound waves into nerve impulses.                                                                                                                                 |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                                                                          |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                                                                   |
| <b>Step 3 — Explicit Reasoning</b> | Sound vibrations move cochlear fluid causing movement of the basilar membrane. Hair cells in the organ of Corti bend, generating receptor potentials that trigger impulses in the auditory nerve. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                                                                         |
| <b>Step 5 — Final Answer</b>       | Sound vibrations move cochlear fluid causing movement of the basilar membrane. Hair cells in the organ of Corti bend, generating receptor potentials that trigger impulses in the auditory nerve. |

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| <b>Common Error</b>  | Students often confuse neural impulses with hormonal responses.  |
| <b>Exam Strategy</b> | Identify stimulus → pathway → effector → physiological response. |

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| <b>Question</b>                    | A graph shows rising thyroxine levels causing reduced TSH secretion. Explain.                                             |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                  |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                           |
| <b>Step 3 — Explicit Reasoning</b> | Increased thyroxine inhibits hypothalamus and pituitary activity, reducing TRH and TSH release through negative feedback. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                 |
| <b>Step 5 — Final Answer</b>       | Increased thyroxine inhibits hypothalamus and pituitary activity, reducing TRH and TSH release through negative feedback. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                           |
| <b>Exam Strategy</b>               | Identify stimulus → pathway → effector → physiological response.                                                          |

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| <b>Question</b>                    | Explain how insulin lowers blood glucose after a meal.                                                                                                |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                              |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                       |
| <b>Step 3 — Explicit Reasoning</b> | High blood glucose stimulates pancreatic beta cells to release insulin. Insulin promotes glucose uptake by cells and glycogen synthesis in the liver. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                             |
| <b>Step 5 — Final Answer</b>       | High blood glucose stimulates pancreatic beta cells to release insulin. Insulin promotes glucose uptake by cells and glycogen synthesis in the liver. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                                                       |

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| <b>Exam Strategy</b> | Identify stimulus → pathway → effector → physiological response. |
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| <b>Question</b>                    | Explain how glucagon raises blood glucose during fasting.                                                                                                |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                                 |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                                          |
| <b>Step 3 — Explicit Reasoning</b> | When blood glucose falls, pancreatic alpha cells release glucagon. Glucagon stimulates glycogen breakdown in the liver releasing glucose into the blood. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                                |
| <b>Step 5 — Final Answer</b>       | When blood glucose falls, pancreatic alpha cells release glucagon. Glucagon stimulates glycogen breakdown in the liver releasing glucose into the blood. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                                                          |
| <b>Exam Strategy</b>               | Identify stimulus → pathway → effector → physiological response.                                                                                         |

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| <b>Question</b>                    | Explain how the hypothalamus links nervous and endocrine systems.                                                                        |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                                 |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                          |
| <b>Step 3 — Explicit Reasoning</b> | The hypothalamus receives neural signals and releases hormones that control pituitary secretion, linking neural and hormonal regulation. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                                |
| <b>Step 5 — Final Answer</b>       | The hypothalamus receives neural signals and releases hormones that control pituitary secretion, linking neural and hormonal regulation. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                                          |
| <b>Exam Strategy</b>               | Identify stimulus → pathway → effector → physiological response.                                                                         |

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| <b>Question</b>                    | A toxin blocks acetylcholine receptors at neuromuscular junctions. Predict the effect.                                              |
| <b>Step 1 — Identification</b>     | Identify the biological structures or hormones involved.                                                                            |
| <b>Step 2 — Principle</b>          | Apply relevant nervous or endocrine physiology.                                                                                     |
| <b>Step 3 — Explicit Reasoning</b> | If acetylcholine cannot bind to receptors, the muscle fiber cannot depolarize and contraction cannot occur, resulting in paralysis. |
| <b>Step 4 — Validation</b>         | Confirm explanation aligns with physiological mechanisms.                                                                           |
| <b>Step 5 — Final Answer</b>       | If acetylcholine cannot bind to receptors, the muscle fiber cannot depolarize and contraction cannot occur, resulting in paralysis. |
| <b>Common Error</b>                | Students often confuse neural impulses with hormonal responses.                                                                     |
| <b>Exam Strategy</b>               | Identify stimulus → pathway → effector → physiological response.                                                                    |